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17/675,340	02/18/2022	Yu Fang	SONA-0006-U01-C03	9356

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EXAMINER	
ROY, SANJOY K	

ART UNIT	PAPER NUMBER
2443	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/675,340	Applicant(s) Fang et al.	
	Examiner SANJOY K ROY	Art Unit 2443	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 9/221/2020.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-21 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-21 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 02/18/2022, 02/18/2022.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. Claims 1-21 have been examined and rejected.

Double Patenting

3 The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the claims at issue are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., In re Berg, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); In re Goodman, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); In re Longi, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); In re Van Ornum, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); In re Vogel, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and In re Thorington, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on a nonstatutory double patenting ground provided the reference application or patent either is shown to

be commonly owned with this application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO internet Web site contains terminal disclaimer forms which may be used. Please visit http://www.uspto.gov/forms/. The filing date of the application will determine what form should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to http://www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

3. Claims 1-21 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-21 of co-pending U.S. Application No. 17570125. Although the claims at issue are not identical, they are not patentably distinct from each other because claims in the co-pending application anticipated claims in the present application.

Instant Application 17675340	US Application No. 17570125
1. A system comprising: a vehicle having a first network and a second network of a different type than the first network; and a converged network device (CND) interposed between the first network and the second	1. A system comprising: a vehicle having a first network and a second network of a different type than the first network; and a communication device interposed between

network and comprising a plurality of ports on the first network; wherein: the CND facilitates communications between the first and the second networks; and a first port of the plurality mirrors a second port of the plurality.	the first network and the second network and structured to: generate network status data by monitoring communications of one or more devices on the first and/or the second networks; and transmit the network status data to a sensor device external to the vehicle. 2. The system of claim 1, wherein: the communication device comprises a plurality of ports on the first network; and the communication device is further structured to configure at least a first port of the plurality to mirror at least a second port of the plurality.

The scope of independent Claims 13 and19 of the instant application is covered by claims 13 and 19 of the co-pending application 17570125.

The scope of the dependent claims can Claims 2-12 of the instant application is covered by claims 2-12 of the co-pending application.

Claim Rejections - 35 USC § 103

4. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

5. Claims 1-21 are rejected under AIA 35 U.S.C. 103 as being unpatentable over Chae et al. (U.S. PGPub 2019/0079842) in view of Kim et al. (U.S. PGPub 2018/0072250).

As per claims 1, 13 and 19,

Chae teaches a system comprising: a vehicle having a first network and a second network of a different type than the first network (Chae see para 0063, 0069 the vehicle 10 an Ethernet terminal 25 part of ethernet network as the first network connected to controller area network CAN 22 as the second network through a gateway 24, vehicle 10 include different-type in-vehicle networks for connection between the input-output device 25 and the at least one vehicle gateway 24, the connection between at least one of the Ethernet terminal 28 and the telematics device 30 and at least one vehicle gateway 24 may be implemented based on an Ethernet communication scheme., the connection between the OBD terminal 26 as the input-output device and the at least one gate way 24 may be implemented based on a CAN communication method different from the Ethernet communication scheme),

and a converged network device (CND) interposed between the first network and the second network and comprising a plurality of ports on the first network (Chea see para 0070, 0112, plurality of controllers 22 can communicate with other in-vehicle devices connected on the in-vehicle network via an in-vehicle gateway 24 representing a converged network deice CND, in-vehicle gateway 24 can perform switching functions

between a plurality of in-vehicle networks including a CAN and an ethernet network, as shown in fig 2, the in-vehicle gateway 24 may include an Ethernet switch, as shown in fig. 10, a diagnostic device 20 transmit a diagnostic message having an Ethernet communication format to an controller 22_1 via a telematics device 30 or an Ethernet terminal/port 28 where the controller 22_1 can be directly connected to the in-vehicle gateway 24 via an Ethernet communication scheme using a port, as shown in fig. 5a, b reprogramming including remote reprogramming or Ethernet reprogramming via a separate Ethernet port);

wherein: the CND facilitates communications between the first and the second networks (Chea see para 0070, 0072, plurality of controllers 22 can communicate with other in-vehicle devices connected on the in-vehicle network via an in-vehicle gateway 24 representing a converged network deice CND, in-vehicle gateway 24 can perform switching functions between a plurality of in-vehicle networks including a CAN and an ethernet network, the in-vehicle gateway 24 configured to transmit messages and signals according to the diagnosis protocol DoIP for networking with an external diagnosis server 20 and an diagnostic device 20 on the basis of Ethernet/IP network scheme, in-vehicle gateway 24 can provide scheduling and protocol conversion functions for interworking with controllers based on CAN, LIN, FlexRay, as shown in fig 2, the in-vehicle gateway 24 may include an Ethernet switch);

Chea fails to exclusively teach and a first port of the plurality mirrors a second port of the plurality.

In a similar field of endeavor Kim teaches and a first port of the plurality mirrors a second port of the plurality(KIM see para 0074, 0121, gateway 500 representing the

converged network device CND is connected to a CAN-based vehicle network 20 include a plurality of CAN communication channels 530, plurality of CAN communication channels 530 may be connected to the controller 510, fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30 In a step S830, when the verification of the security authentication request message is completed, the controller 510 may transmit a control command to the PHY layer 520, PHY layer 520 may activate the second port P2 connected to the target communication node 600 under control of the controller 510, step S962, the PHY layer 520 may transmit the diagnostic request message to the target communication node 600 through the second port P2 by mirroring, the PHY layer 520 may transmit the diagnostic request message to a plurality of communication nodes through a plurality of ports, target communication node 600 and the external diagnostic apparatus 10 may make a unicast address be included in a message generated after receiving the one-side forwarding request message, communication node 600 and the external diagnostic apparatus 10 may configure a destination of the message as a communication node, the message transmitted by the external diagnostic apparatus 10 may be delivered to the target communication node 600. The unicast address may include information that identifies the port to which the communication node, the destination of the message, is connected).

It would have been obvious to one of ordinary skill in the art to before the effective filling date of the claimed invention to combine the teaching of Chea with the teaching of Kim, as doing so would provide an efficient method for diagnosis for a in vehicle network using a gateway managing the connection states of the communication

ports using port mirroring to forward messages can be selectively to the target communication node or the external diagnostic apparatus during the diagnosis process to enhance security in the diagnostic procedure by activating or deactivating the connection states of the communication ports (Kim see para 0028).

As per claims 2, 14 and 20,

Chea in view of Kim teaches the system of claim 1, wherein the CND is structured to, based at least in part on a port selection command value from a sensor device external to the vehicle, assign: which port, of the plurality of ports, is the first port; and/or which port, of the plurality of ports, is the second port (Kim, see para 0092, 0093, step S830, when the verification of the security authentication request message is completed, the controller 510 may transmit a control command to the PHY layer 520. The PHY layer 520 may activate the second port P2 connected to the target communication node 600 under control of the controller 510, step S832, the target communication node 600 may receive the security authentication information from the gateway 500, controller 610 of the target communication node 600 may transmit a control command to the PHY layer 620 to activate the sixth port P6 connected to the gateway 500 under control of the controller 610, communications between the external diagnostic apparatus 10 and the target communication node 600 may be allowed by activating the second port P2 and the sixth port P6).

It would have been obvious to one of ordinary skill in the art to before the effective filling date of the claimed invention to combine the teaching of Chea with the

teaching of Kim, and the motivation to combine the teachings will be the same a stated above for the motivation with relation to claims 1, 13 and 19;

As per claims 3 and 15,

Chea in view of Kim teaches the system of claim 1, wherein the first port of the plurality mirrors a third port of the plurality (Kim see para 0109 when there are a plurality of target communication nodes, a message received through the first port P1 may be mirrored so as to be transmitted through a plurality of ports p2 and p3).

It would have been obvious to one of ordinary skill in the art to before the effective filling date of the claimed invention to combine the teaching of Chea with the teaching of Kim, and the motivation to combine the teachings will be the same a stated above for the motivation with relation to claims 1 and 13;

As per claim 4,

Chea in view of Kim teaches the system of claim 1 further comprising: a sensor device external to the vehicle and structured to monitor one or more devices on the first and/or the second networks via communicatively coupling with the first port(Kim see para 0121, In a step S1050, the external diagnostic apparatus 10 may transmit a diagnostic request message and service time information. The diagnostic request message may include a unicast address indicating the target communication node 600, PHY layer 520 may receive the diagnostic request message and service time information, forward the diagnostic request message and the service time information to the controller 510, controller 510 may determine a port ID of the second port P2 to

which the target communication node 600, which is the destination of the message, is connected from the unicast address included in the diagnostic request message, the unicast address may include an index key that identifies the destination of the message, controller 510 may determine the communication node that matches the index key and determine an ID of a port to which the message is to be forwarded, controller 510 may determine the port ID of the port to which the diagnostic request message is to be transmitted from the unicast address included in the diagnostic request message, external diagnostic apparatus 10 intends to diagnose an electrical component in the vehicle, the external diagnostic apparatus 10 may transmit a diagnostic request message to the gateway 500).

It would have been obvious to one of ordinary skill in the art to before the effective filling date of the claimed invention to combine the teaching of Chea with the teaching of Kim, and the motivation to combine the teachings will be the same a stated above for the motivation with relation to claims 1 and 13;

As per claim 5,

Chea in view of Kim teaches the system of claim 1 further comprising: a sensor device external to the vehicle and structured to monitor a flow on the first and/or the second networks via communicatively coupling with the first port (Kim, see para 0092, 0093, step S830, when the verification of the security authentication request message is completed, the controller 510 may transmit a control command to the PHY layer 520. The PHY layer 520 may activate the second port P2 connected to the target communication node 600 under control of the controller 510, step S832, the target

communication node 600 may receive the security authentication information from the gateway 500, controller 610 of the target communication node 600 may transmit a control command to the PHY layer 620 to activate the sixth port P6 connected to the gateway 500 under control of the controller 610, communications between the external diagnostic apparatus 10 and the target communication node 600 may be allowed by activating the second port P2 and the sixth port P6).

It would have been obvious to one of ordinary skill in the art to before the effective filling date of the claimed invention to combine the teaching of Chea with the teaching of Kim, and the motivation to combine the teachings will be the same a stated above for the motivation with relation to claims 1 and 13;

As per claim 6,

Chea in view of Kim teaches the system of claim 1, wherein the CND is further structured to: assign a first device, of a plurality of devices on the second network, to a first assigned port of the plurality of ports; identify communications data on the second network corresponding to the first device; and transmit the identified communications data corresponding to the first device on the first network via the first assigned port

(Kim see para 0121, In a step S1050, the external diagnostic apparatus 10 may transmit a diagnostic request message and service time information. The diagnostic request message may include a unicast address indicating the target communication node 600, PHY layer 520 may receive the diagnostic request message and service time information, forward the diagnostic request message and the service time information to the controller 510, controller 510 may determine a port ID of the second

port P2 to which the target communication node 600, which is the destination of the message, is connected from the unicast address included in the diagnostic request message, the unicast address may include an index key that identifies the destination of the message, controller 510 may determine the communication node that matches the index key and determine an ID of a port to which the message is to be forwarded, controller 510 may determine the port ID of the port to which the diagnostic request message is to be transmitted from the unicast address included in the diagnostic request message, external diagnostic apparatus 10 intends to diagnose an electrical component in the vehicle, the external diagnostic apparatus 10 may transmit a diagnostic request message to the gateway 500);

It would have been obvious to one of ordinary skill in the art to before the effective filling date of the claimed invention to combine the teaching of Chea with the teaching of Kim, and the motivation to combine the teachings will be the same a stated above for the motivation with relation to claims 1 and 13;

As per claims 7, 15 and 21,

Chea in view of Kim teaches system of claim 6, wherein the CND is further structured to: assign a second device, of the plurality of devices on the second network, to a second assigned port of the plurality of ports; identify communications data on the second network corresponding to the second device; and transmit the identified communications data corresponding to the second device on the first network via the second assigned port (Kim see para 0121, In a step S1050, the external diagnostic apparatus 10 may transmit a diagnostic request message and service time information.

The diagnostic request message may include a unicast address indicating the target communication node 600, PHY layer 520 may receive the diagnostic request message and service time information, forward the diagnostic request message and the service time information to the controller 510, controller 510 may determine a port ID of the second port P2 to which the target communication node 600, which is the destination of the message, is connected from the unicast address included in the diagnostic request message, the unicast address may include an index key that identifies the destination of the message, controller 510 may determine the communication node that matches the index key and determine an ID of a port to which the message is to be forwarded, controller 510 may determine the port ID of the port to which the diagnostic request message is to be transmitted from the unicast address included in the diagnostic request message, external diagnostic apparatus 10 intends to diagnose an electrical component in the vehicle, the external diagnostic apparatus 10 may transmit a diagnostic request message to the gateway 500);

It would have been obvious to one of ordinary skill in the art to before the effective filling date of the claimed invention to combine the teaching of Chea with the teaching of Kim, and the motivation to combine the teachings will be the same a stated above for the motivation with relation to claims 1 and 13;

As per claim 8,

Chea in view of Kim teaches the system of claim 6, wherein the CND is further structured to: assign a second device, of the plurality of devices on the second network, to the first assigned port; identify communications data on the second network

corresponding to the second device; and transmit the identified communications data corresponding to the second device on the first network via the first assigned port(Kim see para 0121, In a step S1050, the external diagnostic apparatus 10 may transmit a diagnostic request message and service time information. The diagnostic request message may include a unicast address indicating the target communication node 600, PHY layer 520 may receive the diagnostic request message and service time information, forward the diagnostic request message and the service time information to the controller 510, controller 510 may determine a port ID of the second port P2 to which the target communication node 600, which is the destination of the message, is connected from the unicast address included in the diagnostic request message, the unicast address may include an index key that identifies the destination of the message, controller 510 may determine the communication node that matches the index key and determine an ID of a port to which the message is to be forwarded, controller 510 may determine the port ID of the port to which the diagnostic request message is to be transmitted from the unicast address included in the diagnostic request message, external diagnostic apparatus 10 intends to diagnose an electrical component in the vehicle, the external diagnostic apparatus 10 may transmit a diagnostic request message to the gateway 500);

It would have been obvious to one of ordinary skill in the art to before the effective filling date of the claimed invention to combine the teaching of Chea with the teaching of Kim, and the motivation to combine the teachings will be the same a stated above for the motivation with relation to claims 1 and 13;

As per claim 9,

Chea in view of Kim teaches the system of claim 1, wherein the CND further comprises an Ethernet switch that includes the plurality of ports(Kim see para 0054,0075, gateway 100 is connected with switches 110, 110-1, 110-2, 120, and 130, and supports a switch which supports an Ethernet protocol, and control at least one of the end nodes 111, 112, 113, 121, 122, 123, 131, 132, and 133 connected to each of the switches where of the end nodes 111, 112, 113, 121, 122, 123, 131, 132, and 133 may include an electronic control unit ECU configured to control various types of devices mounted within a vehicle, gateway 500 include a plurality of communication ports P1, P2, P3, P4, and P5, first port P1 may be connected to an external diagnostic apparatus 1, second to fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30);

It would have been obvious to one of ordinary skill in the art to before the effective filling date of the claimed invention to combine the teaching of Chea with the teaching of Kim, and the motivation to combine the teachings will be the same a stated above for the motivation with relation to claims 1 and 13;

As per claims 10 and 16,

Chea in view of Kim teaches the system of claim 1, wherein the first network is Ethernet based (Chea see para 0070, 0112, plurality of controllers 22 can communicate with other in-vehicle devices connected on the in-vehicle network via an in-vehicle gateway 24 representing a converged network deice CND, in-vehicle gateway 24 can

perform switching functions between a plurality of in-vehicle networks including a CAN and an ethernet network, as shown in fig 2, the in-vehicle gateway 24 may include an Ethernet switch, as shown in fig. 10, a diagnostic device 20 transmit a diagnostic message having an Ethernet communication format to an controller 22_1 via a telematics device 30 or an Ethernet terminal/port 28 where the controller 22_1 can be directly connected to the in-vehicle gateway 24 via an Ethernet communication scheme using a port, as shown in fig. 5a, b reprogramming including remote reprogramming or Ethernet reprogramming via a separate Ethernet port);

As per claims 11 and 18,

Chea in view of Kim teaches the system of claim 1, wherein the second network is controller area network (CAN) based Chea see para 0070, 0112, plurality of controllers 22 can communicate with other in-vehicle devices connected on the in-vehicle network via an in-vehicle gateway 24 representing a converged network deice CND, in-vehicle gateway 24 can perform switching functions between a plurality of in-vehicle networks including a CAN and an ethernet network, as shown in fig 2, the in-vehicle gateway 24 may include an Ethernet switch, as shown in fig. 10, a diagnostic device 20 transmit a diagnostic message having an Ethernet communication format to an controller 22_1 via a telematics device 30 or an Ethernet terminal/port 28 where the controller 22_1 can be directly connected to the in-vehicle gateway 24 via an Ethernet communication scheme using a port, as shown in fig. 5a, b reprogramming including remote reprogramming or Ethernet reprogramming via a separate Ethernet port).

As per claims 12 and 18,

Chea in view of Kim teaches the system of claim 1, wherein the first network is Ethernet based and the second network is controller area network (CAN) based (Chea see para 0070, 0112, plurality of controllers 22 can communicate with other in-vehicle devices connected on the in-vehicle network via an in-vehicle gateway 24 representing a converged network deice CND, in-vehicle gateway 24 can perform switching functions between a plurality of in-vehicle networks including a CAN and an ethernet network, as shown in fig 2, the in-vehicle gateway 24 may include an Ethernet switch, as shown in fig. 10, a diagnostic device 20 transmit a diagnostic message having an Ethernet communication format to an controller 22_1 via a telematics device 30 or an Ethernet terminal/port 28 where the controller 22_1 can be directly connected to the in-vehicle gateway 24 via an Ethernet communication scheme using a port, as shown in fig. 5a, b reprogramming including remote reprogramming or Ethernet reprogramming via a separate Ethernet port);

Conclusion

6. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. This includes:

U.S. PGPub 2018/0286152 Apparatus and method for converting protocol by type of data and vehicle system.

U.S. PGPub 2019/0020985 System and method for wireless interface selection and for communication and access control of subsystems, devices, and data in a vehicular environment

U.S. PGPub 2018/0287815 which describes an interface for connecting ethernet network with CAN network.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to examiner Sanjoy Roy, whose telephone number is 571-270-0675. The examiner can normally be reached on Mon-Fri, 8am.-5pm. (EST).

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Nicholas Taylor can be reached on 571-272-3889. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/SANJOY ROY/
Examiner, Art Unit 2443

/NICHOLAS R TAYLOR/
Supervisory Patent Examiner, Art Unit 2443